2.7 Solving Multi-Step, Real-World Problems With Fractions or Decimals

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• How can you tell what steps and operations are needed to solve three-step problems?
• How do you know if your answer to a multi-step problem is reasonable?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, students continue to solve real-world, multi-step word problems involving rational numbers, using the problem-solving skills they developed in earlier lessons from this unit. Students are guided through problem solving with three-step real-world problems involving decimals and fractions. In the Collaboration, students specifically apply these skills to analyze the distance between hurdles in a 110-meter hurdle race, then determine which arithmetic operations are relevant and use them to solve the problems. Additionally, students draw or use a diagram to help them make sense of measurements, as well as to communicate their reasoning about the measurements.</td></tr>
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<u>Additional Terms</u>: N/A</td><td>• (Optional) Chart paper
• (Optional) Markers</td><td>• If you plan to go to the school track, prepare hurdles or something that will mimic the hurdles and measuring tools.</td></tr>
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Teacher Tips

Common Misconceptions	Things to Consider
• Students may create inaccurate numeric expressions. • Students may make computational errors when performing operations on multi-digit decimal and fraction problems. • Students may have difficulty identifying the steps needed to solve problems, and in which order to solve the steps to find the correct solution. • In 2.7.3 Collaboration, students may create 10 spaces (instead of 9) between 10 hurdles, leading to a miscalculation of the distance between hurdles and incorrect division in the following steps.	• To foster flexible thinking and problem-solving strategies, ask students to draw a visual model first and then create a numeric expression from the visual model. • Provide students who may need assistance organizing information a graphic organizer and/or annotation strategies while engaging with three-step problems. • For the context of the 110-meter hurdles in 2.7.3 Collaboration, the distance between the first and last hurdles is 82.26 meters. If students end up with a non-terminating decimal, have them to revisit each step to find the error. • If time is an issue, then consider assigning 2.7.4 Your Turn! for homework.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share Utilize Think-Pair-Share opportunities in 2.7.2 Guided Instruction and 2.7.4 Your Turn! for students to practice articulating their strategies and approaches in problem-solving in order to continue building fluency in real-world contexts.	**MA.K12.MTR.7.1: Real-World Contexts** Throughout the lesson, students will be reading, analyzing, and discussing real-world scenarios to determine the appropriate mathematical operation to apply for solving problems. Students use concrete and abstract models to investigate real-world problems and find solutions that are reasonable given the context.

Differentiate Instruction

Consider using a jigsaw for 2.7.4 Your Turn! practice problems. Students can be assigned to work on one of the five problems, then rotate to new groups where one student is responsible for sharing thinking and answers for the problem as the other students work on it.

Reading comprehension is at the forefront of learning for real-world problems.

Consider your students when planning for 2.7.5 Wrap-Up – this might work for your students if they need independent processing time, or this activity could be transformed into a Socratic seminar-style debate with evidence required in collaboration with an English teacher.

Lesson Notes:

2.7.1 Warm-Up: Short Division

Component Overview: *Students investigate another method of division, short division.*

2.7.1 Warm-Up: Short Division

In Mexico, when dividing by a single-digit divisor, sometimes a method called "short division" is used. Consider the example shown of $74 \div 4$.

$$\begin{array}{r} 18 \\ 4\overline{)\,7{}^{3}4_{2}} \end{array}$$

1. Using your knowledge of division, explain where you think the 1, 3, 8, and 2 came from in the process.
   First, the tens place is divided. 40 divides into 70 only 1 time so that is where the 1 comes from. 3 represents the remainder 30. Next the ones place is divided. 4 goes into 34 8 times and has a remainder of 2.

2. Show how you might represent the same division problem using a method different from "short division."
   Answers will vary.

Clarify, Correct, Critique
Students analyze a new method of division, short division. To strengthen metacognitive thinking and problem-solving skills, consider positioning students to clarify, correct, and/or critique their own and others' reasoning by analyzing the new method, and comparing it to the standard long division algorithm.

2.7.2 Guided Instruction: Three-Step Rational Problems

Component Overview: *Students practice planning and solving three-step rational problems. Encourage students to recall strategies used to solve one- and two-step real world problems in previous lessons to build upon background knowledge. Whole group and or small group work can be incorporated in this activity to provide differentiation for different types of learners.*

2.7.2 Guided Instruction: Three-Step Rational Problems

1. For his recital, a student practiced the piano 2.18 hours each day for 5 days last week. This week, he practiced 0.75 hours each day for 4 days. How many hours did he spend practicing? 13.9 *hours*

2. The Across Florida 200 Trail Run is an event where participants run 200 miles of trails between the Withlacoochee Bay Trail and Crescent Beach. To prepare for the Across Florida 200 Trail Run, Walt ran 2.18 hours each day for 5 days last week. This week, he ran 2.75 hours each day for 4 days. Walt runs at an average pace of 6.23 miles per hour.
   a. How many hours has Walt run over the last two weeks? 21.9 *hours*
   b. How many miles has Walt run over the last two weeks? 136.437 *miles*
   c. Based on Walt's average pace, how much longer would it take him to run the Across Florida 200 Trail Run than the amount of time he has spent practicing the last two weeks? $\approx$ 10.2 *hours longer*

3. The Across Florida 200 Trail Run offers a team competition where a team of 4 runners take turns to run the 200 miles.
   A team of runners and their average pace is shown.

Name	Molly	Reginald	Tyler	Liz
Average Pace (miles per hour)	6.75	6.75	8.2	9.58

To prepare for the race the team ran part of the trail.

Team Member	**Description**	**Miles**
Tyler	Withlacoochee Bay Trail to Lake Rousseau Dam & Recreational Area	10
Liz	Lake Rousseau to Florida Trail	15
Molly	Florida Trail Part 1	35
Reginald	Florida Trail Part 2	25

How long did it take the team to run the practice run of 85 miles? 11.7 *hours*

Think-Pair-Share
Providing students with ample processing time using Think-Pair-Share strategy gives both independent and verbal processing for more precise and confident understanding.

Real-world problems require reading and comprehension. Consider pairing with a reading teacher for interdisciplinary support. Students can get additional support in identifying key features of the text that will lead to better comprehension, in addition to more practice time. For example, if students use a specific reading comprehension strategy with another teacher, consider having students apply that strategy in these contexts.

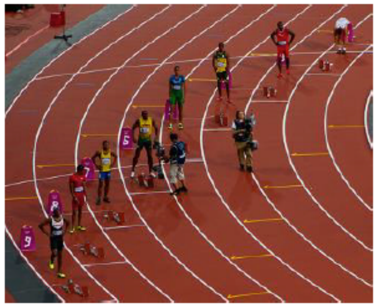
Consider prompting questions to guide students toward key features of the text.
- *What is the question asking? (Total time ran by all)*
- *How will we know how long they took? (Add up how long each person took.)*
- *How do we know how long each person will take? (Multiply their pace times total miles run.)*

2.7.3 Collaboration: Distance Between Two Hurdles

Component Overview: *Students collaborate to analyze and answer questions about the distance between hurdles in a 110-meter hurdle race. Prior to starting the activity, consider showing a video and/or images of 110-meter hurdle races and explain to students that the starting blocks are placed so that the outer and inner lanes still have the same running distance. Consider placing students in small groups based on learning styles and encourage visual and kinesthetic learners to act out or create a model of the scenario. Students then use these models to make sense of the measurements found and communicate their reasoning about the measurements.*

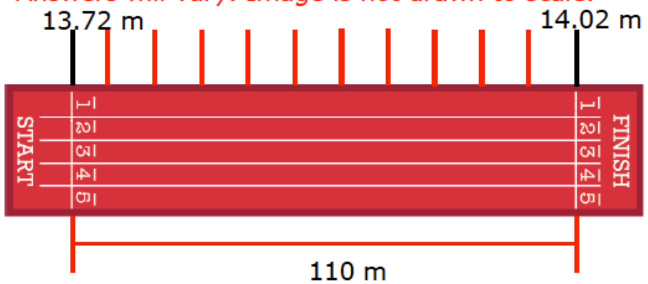
2.7.3 Collaboration: Distance Between Two Hurdles

There are 10 equally spaced hurdles on a racetrack. The first hurdle is 13.72 meters from the start line. The final hurdle is 14.02 meters from the finish line. The racetrack is 110 meters long.



The diagram below represents what the track would look like if the track were straight and not curved.

1. Use the diagram below to show the placement of the hurdles on the racetrack. Label all known measurements.
   Answers will vary. Image is not drawn to scale.



2. Discuss with your partner how many spaces there are between the 10 hurdles.
   9 spaces

3. How far are the hurdles from one another? Explain or show your reasoning.
   $110 - 13.72 - 14.02 = 82.26$
   $82.26 \div 9 = 9.14$
   9.14 *meters*

Before students begin, consider discussing how the hurdles are staggered to compensate for the inner lanes being a shorter distance than the outer lanes. Also, discuss why there are 9 spaces between the 10 hurdles.

Do you have track student-athletes in our class, or do you have a track at the school? Consider taking a field trip and implementing this lesson on your school's track outside to activate visual and kinesthetic entry-points.

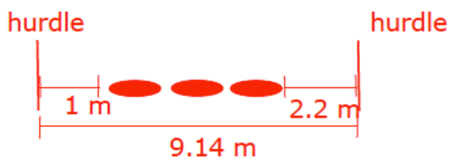
For questions 2 and 3, if students make arithmetic errors when calculating the distance between the first and last hurdles (82.26 meters), ask students to check their work with questions like:

- *How many spaces are needed between 2 hurdles? 3 hurdles? 4 hurdles? How can you extend the pattern to 10 hurdles?*
- *Is your answer reasonable? How can you use estimation and rounding to check?*

2.7.3 Collaboration: Distance Between Two Hurdles (continued)

4. A professional runner takes 3 strides between each pair of hurdles. The runner leaves the ground 2.2 meters before the hurdle and returns to the ground 1 meter after the hurdle.

   a. Draw a diagram to model the runner's strides between the hurdles. Label all known measurements.
   Answers will vary. Image not drawn to scale.



   b. About how long are each of the runner's strides between the hurdles? Show your reasoning.
   $9.14 - 1 - 2.2 = 5.94$
   $5.94 \div 3 = 1.98$
   1.98 *meters*

Clarify, Correct, Critique
As collaborating groups finish and have a completed diagram for question 4a and answer for question 4b, consider combining groups and positioning students to clarify, correct, and/or critique the responses of others. Groups can take turns sharing their diagrams, then the other students can ask clarifying questions, offer suggestions for correction, and critique the diagram for accuracy.

2.7.4 Your Turn!

Component Overview: *The unit has used scaffolding to help students become successful at solving problems using rational numbers. Fluency with rational values is an important focus for the 6^{th} grade accelerated course. Teachers may consider allowing students to use a calculator to answer the questions.*

2.7.4 Your Turn!

1. A pool that holds 273.68 gallons of water lost 4.79 gallons of water each day while there was a leak that went unnoticed. Tad noticed the water was lower, so he added 20.45 gallons of water before he realized there was a leak on the side. If Tad fixed the leak after 8 days of leaking, how many gallons of water does he currently have in the pool?
   255.81 *gallons*

2. A grocery store opens a new wing bar that charges by the pound. Yasmine wants to purchase $\frac{1}{4}$ pound of garlic parmesan wings, $\frac{1}{8}$ pound of mild wings, and $\frac{2}{5}$ pound of teriyaki wings. She also wants to purchase some hot wings. If Yasmine wants to purchase a total of 2 pounds of wings, how many pounds of hot wings should she buy?
   $1\frac{9}{40}$ *pounds*

For students who finish early, challenge them to determine after how many days the pool would be empty if the leak was not fixed (58 days), and to write an expression that could represent that value.

Did you subtract the amount of each type of wing from 2 pounds? Or did you add up all types before subtracting that value from 2 pounds? Would you do this differently next time? Why or why not?

Differentiate the content to engage more learners by giving students the opportunity to substitute one of the questions with their own question and self-made context. They must write the question as well as verify the values chosen by finding the solution.

2.7.5 Wrap-Up: Cool Down

Component Overview: *This Wrap-Up assesses students' place value and arithmetic reasoning in relation to rational numbers and operations. Students record whether they agree, disagree, or are unsure about the statements.*

2.7.5 Wrap-Up: Cool Down

For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	*Answers will vary. Unsure, it depends on if the number is a whole number or a decimal number.*
When you add two positive decimals together, the value is always larger.	*Answers will vary. Agree, the decimal number will grow.*
To add fractions, their denominators must be the same.	*Answers will vary. Agree, to add fractions, they must first be like fractions.*
The product of two fractions is always a larger number.	*Answers will vary. Disagree, when you multiply fractions the product is smaller than the factors.*
When you put a zero on the end of any number, it makes it larger.	*Answers will vary. Unsure, it depends on whether the number is a whole number or a decimal number.*

Differentiating by Product:

- *Link Up: Students write whether they agree or disagree on a whiteboard or use their hand to indicate. Students then must "Link Up" and find someone who does not have the same answer and try to convince them why their answer is correct.*
- *A/S/N: Students must include "Always," "Sometimes," or "Never" in their responses. The first and last prompts will be the "Sometimes" in which students must engage in mathematical discourse.*
- *Prove It: Challenge students to provide evidence for each answer with a justification. Correct answers will only be accepted if there is an example. Counterexamples included will be worth bonus points!*